

- 4 (a) ability to do work B1
as a result of a change of shape of an object/stretched etc B1 [2]
- (b) work = average force $\times$ distance moved (in direction of the force) B1
either work = $\frac{1}{2} \times F \times x$
or work is area under F/x graph which is $\frac{1}{2}Fx$ B1
 $F = kx$ B1
so work / energy = $\frac{1}{2}kx^2$ A0 [3]
- (c) (i) spring constant = $\frac{3.8}{2.1}$ M1
= 1.8 N cm^{-1} A0 [1]
- (ii) 1 $\Delta E_P = mg\Delta h$ or $W\Delta h$ C1
= $3.8 \times 1.5 \times 10^{-2}$
= 0.057 J A1 [2]
2 $\Delta E_S = \frac{1}{2} \times 1.8 \times 10^2 (0.036^2 - 0.021^2)$ M1
= 0.077 J A0 [1]
3 work done = $0.077 - 0.057$
= 0.020 J A1 [1]
(allow e.c.f. if $\Delta E_S > \Delta E_P$)

[Total: 10]

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5 (a) (i) frequency f